3 Gender performances of male and female politicians on social media

A corpus-assisted discourse study

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3.1 Introduction

Language is a powerful tool for influence, and political discourse often reflects and reinforces existing power structures. Gendered language can perpetuate traditional power hierarchies by associating certain words, expressions, or styles of communication with masculine or feminine attributes (Coates, 2015). Research has consistently found that distinct personality traits are associated with men and women, with women typically perceived as warm, gentle, compassionate, and family-oriented, while men are seen as tough, aggressive, assertive, and achievement-oriented (Hayes, 2011; Holmes & Meyerhoff, 2003; Huddy & Terkildsen, 1993; Lee & Lim, 2016).

These stereotypes extend into the political arena, where voters often attribute specific characteristics to politicians based on traditional gender roles (Bigler & Liben, 2007). The expectation is that male leaders will display strength and toughness, whereas women are expected to exhibit warmth and compassion (Carli & Eagly, 2001; Heilman, 2001). This has historically limited women's participation in politics, as public offices tend to prioritize masculine traits, especially for high-ranking positions. Men are perceived as more competent in dealing with "masculine" issues such as economy, military, foreign affairs, technology, science, crime, terrorism, and gun problems; women are seen as more suitable for "feminine" issues like education, health, human rights, women's rights, animal rights, child care, poverty, arts, environment, and social welfare (Huddy & Terkildsen, 1993; Sanbonmatsu, 2002).

To counteract these stereotypes, female politicians often adopt strategies to emphasize their masculine qualities, such as adopting a combative campaign style (Huddy & Terkildsen, 1993; Lee, 2013) or presenting themselves in more formal attire (Trent & Friedenberg, 1991). Female politicians also actively discuss political issues to showcase their expertise (Evans & Clark, 2016). By contrast, male politicians do not seem to feel as strongly compelled to stress masculine traits as women do. Instead, male candidates often clamor to appear sympathetic, kind, and accessible, which are typical feminine traits (Huddy & Terkildsen, 1993).

DOI: 10.4324/9781003449379-5

Therefore, female politicians face a double bind in politics. They are often expected to exhibit traditionally "masculine" qualities such as assertiveness, toughness, and decisiveness to be taken seriously as leaders. However, when they display these characteristics, they can be criticized for not being warm or likable – traits that are stereotypically "feminine" and expected of women. Conversely, if they are too warm, gentle, or emotionally expressive, they risk being seen as weak or not fit for leadership (Burns & Murdie, 2018; Diekman, 2007). Female politicians who concentrate on "women's issues" such as childcare, reproductive rights, and equal pay might strengthen their support among women. However, they may also be pigeonholed or marginalized, with critics suggesting they lack the breadth of knowledge or capability to address "harder" issues like the economy, defense, or foreign policy. If they avoid women's issues, they may be criticized for not advocating for women or for failing to use their unique position to effect change in areas that disproportionately affect women (Evans & Clark, 2016; Herrnson et al., 2003; Jamieson, 1995).

Previous studies have demonstrated that media have the potential to (re) present and reinforce dominant ideologies around gender roles and reproduce the idea that women are inferior to men (Bachmann et al., 2018). Nevertheless, most of these studies focused on traditional media (Everitt et al., 2016). While traditional media have often given more coverage to male politicians, social media presents an opportunity for more equitable representation (Yarchi & Samuel-Azran, 2018; Zhu, 2016). Recent studies have shown that social media can be a supportive environment for female politicians, providing a platform where they can engage more equally with the public (McDonnell, 2020). As the participation of female politicians in government increases and the use of social media in campaigns grows, it is crucial to examine whether social media is mitigating the double bind that female politicians experience (Evans & Clark, 2016; Yarchi & Samuel-Azran, 2018). Therefore, this study combines corpus linguistics and critical discourse analysis to give a corpus-assisted discourse study of the gender performances of male and female politicians on Twitter (X) to examine to what extent their performances are shaped by traditional gender stereotypes. It has two primary objectives: (1) to examine whether the double bind situation still exists on social media for female politicians; (2) to propose a corpus-assisted approach to the examination of gender performances on social media.

3.2 Previous studies on gender and political communication on social media

Current scholarship approaches gender from a social constructionist standpoint, viewing it as an evolving enactment rather than a static, preassigned category (Holmes & Meyerhoff, 2003). This perspective casts gender as a series of interactions and behaviors ("doing") rather than simply an identity or status ("being") (Coates, 2015), with these behaviors collectively informing

societal norms of masculinity and femininity (Butler, 2006). In other words, gender is not an inherent identity or stable notion but rather something that is produced and made to seem natural through repeated performances within a cultural framework (Butler, 2006). This contributes to the shift from the focus on gendered linguistic variations to "gender performances" or "gender performativity" in multiple fields of study (Butler, 2006; Liu et al., 2022; Melendres & Barea, 2021).

Recent research has delved into the application of social media by politicians as a means to connect with constituents and garner support (Evans & Clark, 2016; Evans et al., 2014; Gainous & Wagner, 2013). Since its inception in 2006, Twitter (X) has grown to be a key communication resource for political figures (Evans et al., 2014), serving as an effective and efficient medium to reach essential demographics (Jackson & Lilleker, 2011). As suggested by Jackson and Lilleker (2011), the use of Twitter (X) by politicians will be underpinned by two interrelated theories: (1) impression management and (2) constituency service. Impression management concerns the motivation to create favorable impressions and the ability to impress the target audience (Leary & Kowalski, 1990), and constituency service concerns the constituency roles that the politicians are expected to serve. They observe that Twitter (X) serves predominantly as a platform for impression management and, to a lesser degree, for constituency service.

Social media platforms like Twitter (X) offer female politicians increased avenues for self-promotion and advancing their position in the political hierarchy (Yarchi & Samuel-Azran, 2018). Given the existing gender imbalances in politics, female politicians may employ distinct strategies when highlighting certain personality traits or issues during campaigns (Lee & Lim, 2016). Compared to their male colleagues, women in politics are more engaged in Twitter (X) campaigning and more likely to address policy and women's issues (Evans & Clark, 2016). They also tend to be more critical of their opponents (Evans et al., 2014).

This has given rise to a number of studies on gender performances of female and male politicians on social media, but different and even ambivalent findings have been found (Lee, 2013; Lee & Lim, 2016; McDonnell, 2020; Meeks, 2016). For example, Yarchi and Samuel-Azran (2018) investigated whether social media levels the playing field for female politicians in terms of engagement and support mobilization, finding no significant differences in rhetoric but rather in the issues addressed between genders. Lee (2013) explores how male and female politicians present themselves online, noting that women often portray themselves as strong leaders by accentuating masculine traits. Meeks (2016) finds that personalization and interactivity on Twitter (X) are employed by both genders, with women being more interactive. Cardo (2021) reviews content from Theresa May, Hillary Clinton, and Jacinda Ardern across Twitter (X) and Facebook to examine their use of digital platforms in feminizing politics during election campaigns, revealing that they employ both masculine and feminine communication styles and

address both hard and soft policy issues. These studies suggest the necessity of a systematic examination of gender performances of male and female politicians on social media to further verify whether the double bind exists on social media. In addition, previous studies used to focus on manual content analysis and on specific politicians' Twitter (X) use, but recent studies are increasingly utilizing computational methods to dissect language use across genders (Windsor et al., 2022).

3.3 From critical discourse analysis to corpus-assisted discourse studies

Critical discourse analysis (CDA) is an analytical approach that explores the relationship between language, power, and society (Wodak & Meyer, 2016). It views discourse as a form of social practice and argues that discourse is socially shaped and socially shaping/constitutive (Fairclough, 1992). It can be constitutive in three aspects: (1) social identities, (2) social relations, and (3) social knowledge and experience. In the context of gender, language is a crucial tool for the expression and reinforcement of power structures. By critically analyzing language, CDA seeks to uncover and critically examine how language contributes to the construction, reinforcement, and dissemination of gender norms, roles, and power dynamics (Baker, 2008; Liu et al., 2022; Wodak, 1997). It has been widely applied to the examination of the discursive constructions of gender in different sociopolitical contexts (Fairclough, 1989; Kotthoff & Wodak, 1997).

CDA used to rely on the qualitative analysis of a small sample of texts, but with the development of corpus linguistics (CL) in the 1990s, a few studies attempted to combine CL and (critical) discourse analysis (Hardt-Mautner, 1995; Stubbs, 1996). This contributed to a growing number of studies toward this endeavor (Baker, 2006; Fairclough, 2000; Flowerdew, 1997). Nevertheless, as Baker et al. (2008) critically reviewed, most of previous studies were conducted by researchers either in (critical) discourse analysis or CL with a meager knowledge of the methods and theories in other disciplines, and most of previous studies featured an imbalanced combination of the methods and theories of both disciplines. Therefore, they argue for a balanced combination or "synergy" of the methods and theories from both disciplines.

Corpus-assisted discourse studies (CADS) proposes to combine the methods and theories of CDA and CL and give a balanced combination of the methods and theories of the two disciplines (Partington et al., 2004). It has been increasingly applied in the investigation of the discursive constructions of different topics, such as migrants and immigrants (Baker et al., 2008), Iraq war (Haarman & Lombardo, 2009; Morley & Bayley, 2009), environment (Liu & Huang, 2023; Liu & Li, 2017; Liu & Zhang, 2018), Hong Kong (Liu & Jiang, 2019; Liu & Lin, 2021; Liu & Ma, 2022; Liu & Zhong, 2020), and Europe (Marchi & Taylor, 2009). It has also been employed in the discursive constructions of gender (Baker, 2005, 2008; Milani, 2013; Wang & Ma,

2020). Nevertheless, most previous studies have relied on the traditional CL methods, such as frequencies, collocates, and keywords. The development of computational linguistics and text mining calls for the incorporation of more sophisticated computer-assisted analytic methods in CADS in order to present a more efficient and accurate analysis of a large sample of texts.

3.4 Data collection and analytic methods

3.4.1 Data collection

Twitter (X), a microblogging and social networking service, enables its users to send and read short 140-character messages known as Tweets (Evans & Clark, 2016). Although launched in 2006, it wasn't until around 2012 that a majority of politicians began to actively use the platform (Evans & Clark, 2016; Meeks, 2016). By December 2022, Twitter (X) boasted over 368 million monthly active users globally (Statista, 2023), establishing itself as a vital channel for political figures to connect instantly with a broad audience, including those beyond the reach of traditional media (Lee & Lim, 2016). For campaign-savvy politicians lacking substantial funds for advertising, Twitter (X) provides a cost-effective means to disseminate their political messages (Evans et al., 2014). Additionally, Twitter (X) has proven to be particularly effective for engaging younger voters, with a significant proportion of daily users being under the age of 30 (Evans et al., 2014). It serves both as a conduit for mass communication and a social network, offering a unique blend of personal and public interaction (McDonnell, 2020).

Previous investigations have shown that women are generally more active on Twitter (X), both in terms of account ownership and frequency of use (Gainous & Wagner, 2013). This study collects the most recent 100 Twitter (X) posts of politicians who are ranked as the top ten most influential political leaders by *Forbes* and *Time* (see Table 3.1). These individuals are all verified on Twitter (X) and regularly update their feeds. Each politician maintains a consistent presence on Twitter (X) and commands a substantial following that reaches into the thousands. The NVivo software suite is employed to collect the latest 100 tweets from these accounts, with a stipulation that only the first tweet of any given day is selected in cases where multiple tweets were posted within a single day. The corpus for female politicians contains 36,229 words, while the one for male politicians includes 37,535 words.

3.4.2 Analytical framework and procedures

This study proposes to examine gender performances of male and female politicians at multilevels of discourse: (1) topics/themes, (2) discursive strategies, and (3) linguistic means and realizations (Wodak, 2016). In order to identify male and female politicians' preferences at the three levels of discourse, different analytic tools and methods have been employed to perform a

Table 3.1 Overview of politicians' Twitter (X) accounts

Name	*Age*	*Gender*	*Country*	*Position*	*Posts*	*Followers*
Kamala Harris	57	F	United States	U.S. vice president	7581	13m
Ursula von der Leyen	63	F	Germany	President of the EU Commission	3485	1.2m
Nirmala Sitharaman	63	F	India	Minister of Finance and Corporate Affairs	20.8k	5m
Nancy Pelosi	82	F	United States	Speaker of the House	13.3k	7.4m
Ngozi Okonjo-Iweala	68	F	Nigeria	Director General of the World Trade Organization	3,369	1.9m
Jacinda Ardern	42	F	New Zealand	Prime Minister of New Zealand	6,919	797.3k
Nicola Sturgeon	52	F	Scotland	First Minister of Scotland	27.7k	1.4m
Zuzana Caputova	49	F	Slovak Republic	President of Slovak Republic	472	180.2k
Theresa May	65	F	United Kingdom	Former Prime Minister of the United Kingdom	1,830	1m
Hillary Clinton	74	F	United States	67th United States Secretary of State	13.5k	31.5m
Ron DeSantis	43	M	United States	46th governor of Florida	5,908	2.5m
Joe Biden	79	M	United States	US president	8,322	34.7m
Kevin McCarthy	57	M	United States	US House Minority Leader	19.9k	1.5m
Narendra Modi	71	M	India	Prime Minister of India	33.5k	80.9m
Mitch McConnell	80	M	United States	U.S. Senate Republican Leader	4,914	2.1m
Cyril Ramaphosa	69	M	Africa	President of the African National Congress	13.2k	2.3m
Justin Trudeau	50	M	Canada	23rd Prime Minister of Canada	40.8k	6.1m
Sadiq Khan	51	M	Britain	Mayor of London	52.1k	1.1m
Chuck Schumer	71	M	United States	US Senate Majority Leader	22.9k	3.4m
Barack Obama	60	M	United States	US president	16.5k	132.3m

top-down analysis of the data and triangulate the findings (Baker et al., 2008; Wodak et al., 2009).

It begins with the identification of key topics/themes used by male and female politicians. The free text mining software KH Coder is used for the analysis of the most prominent topics. KH Coder was first developed for quantitative content analysis, which uses "Stanford POS Tagger to extract words from English data, R for statistical analysis, and MySQL to organize and retrieve the data" (Higuchi, 2016). It is used in this study to produce a co-occurrence network analysis to get an overall picture of the two corpora (Liu et al., 2022). All words generated by the co-occurrence network are words of high frequency that tend to co-occur strongly with one another. Thus the co-occurrence network can display not only words that are distinctive to the corpus but also any close association between words in one corpus. The strongly co-occurring words are connected by solid lines, and they are clustered into different groups differentiated by different colors. In order to identify the topics/themes used by politicians, this study starts with an automatic co-occurrence network analysis of the two corpora. We focused on the top 100 most strongly co-occurring words. All words are filtered by POS and only verbs, proper nouns, and nouns are chosen for visualization because we are concerned about their preferential topics.

It is followed by a close analysis of the top 15 key semantic categories (SMCs) in each corpus when they are compared with each other. The corpus-analytic tool Wmatrix 5.0 is used for key SMC analysis. The corpus annotation tool (USAS) that it incorporates can categorize the tokens in a corpus into 21 semantic fields, which can be further classified into 232 SMCs. Key SMCs refer to those SMCs that are statistically significant in one corpus when compared with a general reference corpus or specialized corpus. In order to show how the two corpora differ in the choice of key SMCs, the two corpora were compared, and two key SMC lists were generated. This study focuses on the top 15 key SMCs. All of these top 15 key SMCs have a log-likelihood value above 6.63 ($p < 0.01$) (L'Hôte, 2010). That means all of them are statistically significant.

This is followed by a close examination of the speech acts adopted by politicians to explore their communication strategies. Speech acts concern what people are doing with words (Austin, 1962). Zhang and Wu (2018) make a distinction between self-oriented and other-oriented speech acts on social media in order to examine how social media users establish and maintain relations on social media (Liu et al., 2022). Self-oriented speech acts consist of three subcategories: (1) promoting self and others, (2) self-reporting mood, (3) self-reporting moment and information. Other-oriented speech acts are further divided into six subcategories: (1) eliciting response, (2) directive, (3) greetings, (4) showing judgment and appreciation, (5) showing gratitude, (6) congratulating. They can be used to examine how male and female politicians interact with their followers on social media. Table 3.2 displays the operation criteria for coding and illustrated examples. Since each post might contain several speech acts simultaneously, this study identifies a maximum of two speech acts for each post.

Table 3.2 Coding scheme for speech acts

Types	*Speech acts*	*Description*	*Examples*
Self-oriented	Promoting self and others	Messages promoting the politicians' or others' achievements, competence, and abilities	*In May, the economy added 390,000 jobs, bringing the total to 8.7 million jobs created since I took office, an all-time record.*
	Self-reporting mood	Messages expressing emotions or inner feelings	*My thoughts are with Australia and the people of Melbourne-such a shocking incident this afternoon. Thinking of all the victims and their families.*
	Self-reporting moment and information	Messages sharing small life stories or personal information.	*Vice President Harris delivers remarks on the Supreme Court decision on Dobbs v. Jackson Women's Health Organization to overturn Roe v. Wade.*
Other-oriented	Eliciting response	Messages eliciting responses from followers or the addressed	*LoveLondon?DogsofTwitter*
	Directive	Messages expressing a request to others, directly or indirectly	*Let us commit ourselves to a future in which all the people of the world can be employed in dignity, safety and freedom and modern slavery becomes a thing of the past.*
	Greetings	Messages conveying daily, seasonal, birthday or festive greetings	*Happy Birthday! Thank you for being the rock for our family I love you very, very much!*
	Showing judgment and appreciation	Messages expressing politicians' reasoning, beliefs or evaluations of people and things	*The cost of living crisis is hammering families, working people and pensioners across London. It's on the Government to step up, not those struggling to make ends meet.*
	Showing gratitude	Messages expressing thankfulness or gratitude to others	*I'm grateful to everyone who has been fighting so hard to make sure people everywhere have an equal opportunity to choose their representatives.*
	Congratulating	Messages congratulating others for their outstanding performance or work	*Congratulations Madame Prime Minister! So proud of you.*

3.5 Data analysis

3.5.1 Co-occurrence network analysis

Figure 3.1 shows the co-occurrence network of male politicians' key topics and themes. **Theme 1** concerns the improvement of people's life and the determination to keep work for people, as can be seen from such words as *people*, *work*, *life*, *continue*, *fight*, *keep*, and *work*. **Theme 2** underlines the need to support Ukraine, as can be seen from such words as *Ukraine*, *support*, *help*, *want*, *support*, *government*, *get*, and *need*. **Theme 3** underlines the inflation problems caused by Democrats, as can be seen from such words as *Americans*, *Democrats*, and *inflation*. **Theme 4** highlights President Biden's border policies, as can be seen from such words as *House*, *border*, *Biden*, and *President*. **Theme 5** discusses creating jobs for people, as can be seen from such words as *create*, *job*, and *nation*. **Theme 6** highlights the energy and gas prices in America, as can be seen from such words as *family*, *price*, *pay*, *gas*, *energy*, and *America*. **Theme 7** underscores the meeting with world leaders, as can be seen from such words as *celebrate*, *world*, *join*, *community*, *leader*, and *meet*. **Theme 8** concerns history, as can be seen from such words

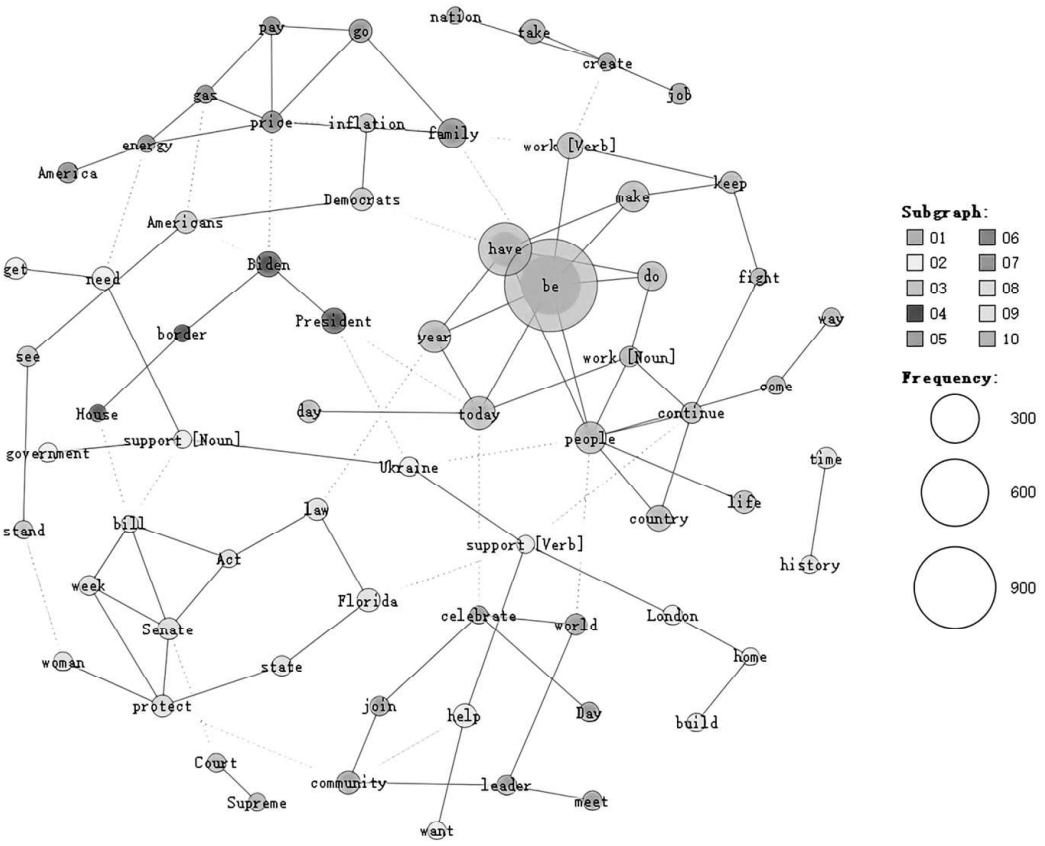


Figure 3.1 Co-occurrence network of male politicians' tweets.

as *history* and *time*. **Theme 9** emphasizes the protection of women, as can be seen from such words as *protect*, *woman*, *Senate*, *bill*, *Act*, *law*, and *state*. **Theme 10** highlights the Supreme Court, as can be seen from the two words *Supreme* and *Court*.

Figure 3.2 displays the co-occurrence network of female politicians' preferred topics/themes. Overall, the strongly co-occurring words are clustered into eight themes. **Theme 1** focuses on the discussion of some international topics with world leaders, such as climate change, energy security, and trade, as can be seen from such words as *discuss*, *leader*, *world*, *meet*, *climate*, *trade*, *security*, and *energy*. **Theme 2** foregrounds the gratitude toward people's work and contribution to the country's future, as can be seen from such words as *work*, *thanks*, *people*, *keep*, *look*, *let*, *country*, and *future*. **Theme 3** highlights the measures taken to control gun violence and the implementation of laws to control national violence, as can be seen from such words as *violence*, *gun*, *law*, *state*, *nation*, *take*, and *step*. **Theme 4** underlines the provision of opportunities for those families in need, as can be seen from such words as *ensure*, *address*, *opportunity*, *work*, *get*, *need*, and *family*. **Theme 5** underlines the EU's support toward Ukraine, as can be seen from such words as *Ukraine*, *Russia*, *war*, *Putin*, *EU*, and *support*. **Theme 6** emphasizes the protection of woman's

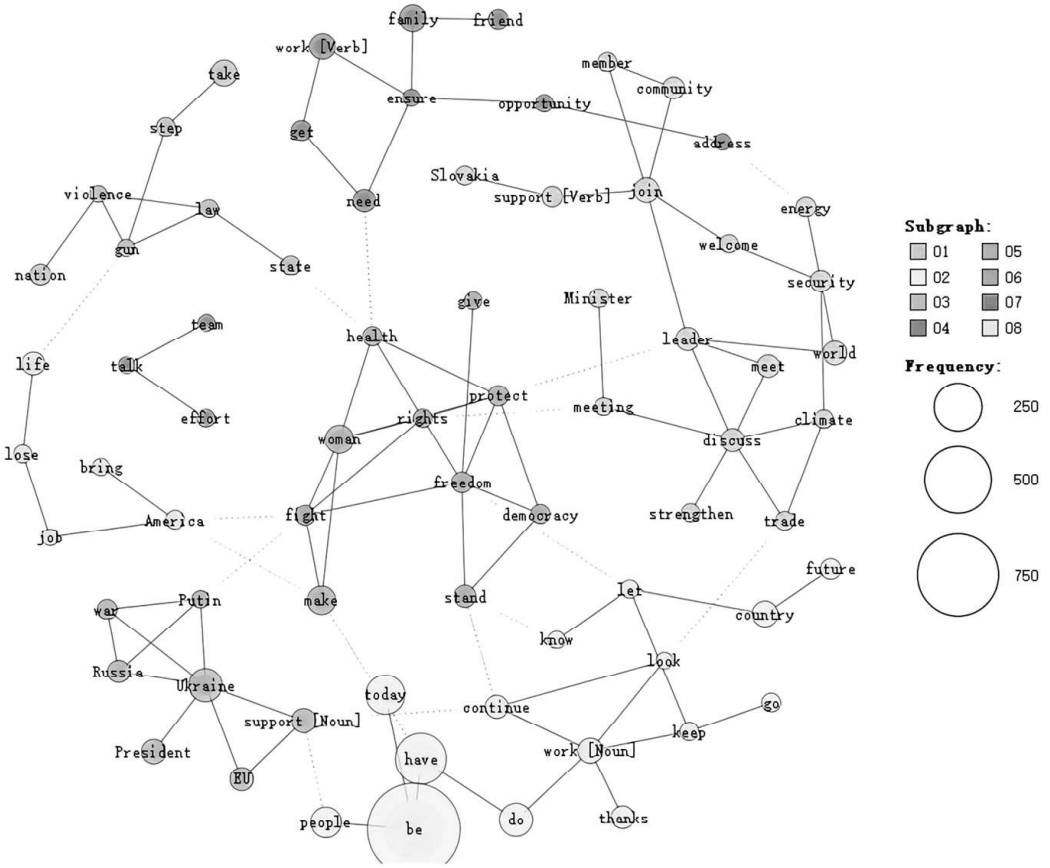


Figure 3.2 Co-occurrence network of female politicians' tweets.

health rights and the fight for freedom and democracy, as can be seen from such words as *woman*, *rights*, *fight*, *protect*, *freedom*, *democracy*. **Theme 7** underlines the importance of cooperation, which can be seen from typical tokens such as *team*, *talk*, *effort*. **Theme 8** is concerned with such topics as job and life losses in America, as can be seen from such words as *America*, *job*, *lose*, and *life*.

Therefore, both male and female politicians foreground some common themes on their social media, such as energy, security, inflation, support toward Ukraine, the protection of women, the creation of jobs, and the contribution to the country's future. Nevertheless, male politicians tend to focus on domestic issues, while female politicians address both international and national issues. Besides, while male politicians focus on internal conflicts between different parties and legislation to solve internal problems, female politicians tend to focus on international collaboration and discussion to address certain international issues. Finally, female politicians tend to care more about so-called women's issues, as can be seen from the central position women's issues occupy (Windsor et al., 2022).

3.5.2 Analysis of key semantic categories

Table 3.3 shows the top 15 key SMCs of each corpus when compared with one another. Previous studies have suggested female language use tends to emphasize collaboration, consensus building, and politeness, while male language use features directness, factualness, and toughness (Duffy & Pruchniewska, 2017; Evans et al., 2014; Windsor et al., 2022). Female politicians are expected to address so-called female issues such as poverty and the environment rather than the "tougher" issues of security and economy (Burns & Murdie, 2018; Diekman, 2007). The same differences can still be detected in their preferences for some key SMCs.

Male politicians tend to focus on those negative and challenging topics that need to be addressed. As Table 3.3 shows, male politicians tend to focus more on money and costs incurred, including **I1.1** ("Money and pay"), **I1.3** ("Money: Cost and price"), **I1.3+** ("Expensive"), and **I1.2** ("Money: Debts"). Their typical tokens are as follows:

- **I1.1**: *funding* (18), *investment* (15), *tax* (15), *taxes* (12), *save* (11), *capital* (9), *investing* (8), *pay* (6)
- **I1.3**: *prices* (38), *bill* (18), *costs* (14), *price* (13), *cost* (10), *bills* (5), *worth* (3), *costing* (2), *expense* (1)
- **I1.3+**: *inflation* (34), *expensive* (1)
- **I1.2**: *pay* (19), *spending* (9), *spent* (8), *loss* (6), *debt* (6), *paying* (6), *spend* (3), *$350_billion* (3), *$65_billion* (2), *owes* (2), *pay_off* (1)

In addition, it focuses on such problems as gas, unemployment, electricity, and drug abuse, including **O1.3** ("Substances and materials: Gas"), **I3.1–** ("Unemployed"), **F3** ("Smoking and nonmedical drugs"), and **O3**

Table 3.3 Top 15 key SMCs in each corpus

Rank	*Female politician*			*Male politician*		
	Item	*LL*	*Semantic category*	*Item*	*LL*	*Semantic category*
1	A6.1+	24.91	Comparing: Similar	I1.1	25.81	Money and pay
2	S2.1	24.29	People: Female	I1.3	18.48	Money: Cost and price
3	S1.2.5+	18.74	Tough/strong	I1.3+	13.76	Expensive
4	I2.2	18.22	Business: Selling	I1.2	13.18	Money: Debts
5	Z4	16.93	Discourse Bin	O1.3	12.82	Substances and materials: Gas
6	Q2.1	14.17	Speech: Communicative	G2.1	12.6	Law and order
7	S1.2.4+	12.42	Polite	I3.1–	12.36	Unemployed
8	X3.2	12.28	Sensory: Sound	O4.4	11.96	Shape
9	A1.4	11.9	Chance, luck	N5–	9.67	Quantities: little
10	S5+	10.74	Belonging to a group	F3	9.44	Smoking and nonmedical drugs
11	Q4.1	10.22	The Media: Books	O3	8.28	Electricity and electrical equipment
12	E3+	9.86	Calm	T1.1.1	7.93	Time: Past
13	Q2.2	7.9	Speech acts	H1	7.91	Architecture, houses, and buildings
14	N2	7.21	Mathematics	A15–	7.76	Danger
15	O4.2	7.13	Judgment of appearance	S1.1.4+	7.42	Deserving

("Electricity and electrical equipment"). The typical tokens in these key SMCs are as follows:

- **O1.3**: *gas* (33), *air* (6), *hydrogen* (1)
- **I3.1–**: *unemployment* (9), *retirement* (3), *retired* (2), *retiring* (1), *on_the_streets* (1), *retires* (1), *strikes* (1)
- **O3**: *circuit* (6), *electric* (4), *battery* (2), *searchlight* (1), *electricity_supply* (1), *electricity* (1)
- **F3**: *cannabis* (4), *traffickers* (2), and *drug_abuse* (1)

Male politicians focus on the implementation of laws and different parties, as can be seen from **G2.1** ("Law and order") and **H1** ("Architecture, houses and buildings").

- **H1:** *house* (19), *building* (18), *build* (16), *rebuild* (5), *memorial* (5), *built* (5), *facility* (4), *bridges* (3)
- **G2.1**: *law* (53), *court* (39), *legislation* (24), *security* (16), *judge* (13), *rule* (12), *laws* (9), *police* (9)

Finally, male politicians look at the negative side, as can be seen from **A15–** ("Danger"), and **N5–** ("Quantities: Little"), and **S1.1.4+** ("Deserving").

A15– underlines the risks and dangers, and **N5–** and **S1.1.4+** suggest insufficiency. Their typical tokens are as follows:

- **A15–:** *dangerous* (6), *risks* (4), *exposure* (3), *risk* (3), *stunt* (3), *endangered* (2), *at_stake* (1), *at_risk* (1)
- **N5–:** *less* (12), and *fewer* (2)

Female politicians tend to construct themselves as strong and tough leaders by highlighting masculine qualities (Lee, 2013), as can be seen from **S1.2.5+** ("Strong"). It includes such tokens as *strong* (34), *strengthen* (14), *strengthening* (13), *resilience* (8), *strength* (8), *strengthened* (2), *tough* (2), *resilient* (2), *tenacious* (1), and *resiliency* (1). In the meantime, female politicians also highlight so-called "tougher issues" of trade, as can be seen from **I2.2**. It consists of such tokens as *trade* (51), *trading* (6), *supply_chains* (4), *consumers* (3), *market* (3), *exports* (3), and *purchased* (2).

Nevertheless, female politicians tend to focus on so-called women issues, including female rights, as can be seen from **S2.1** ("People: Female"). It consists of such tokens as *women* (76), *woman* (17), *Ms.* (6), *madam* (4), *girls* (3), *female* (2), *girl* (1), *lady* (1), *Ms* (1), *Women's* (1), *Madame* (1), *ma'am* (1).

Also, female politicians tend to focus on positive aspects, as can be seen from **A1.4** ("Change, luck"), **E3+** ("Peace"). Instead of focusing on cost and money, female politicians tend to focus on trade. Instead of foregrounding risks and dangers, female politicians highlight opportunities and luck, as can be seen from **A1.4**. Their typical tokens are as follows:

- **A1.4**: *opportunity* (19), *luck* (9), *opportunities* (9), *chance* (6), *coincidence* (2)
- **E3+**: *peace* (10), *rest* (7), *peaceful* (6), *respite* (1), and *reposing* (1)

Further, female politicians tend to focus on collaboration, consensus building, and politeness, as can be seen from **A6.1+** ("Comparing: Similar"), **S5+** ("Belonging to a group), and **S1.2.4+** ("Polite"). Their typical tokens are as follows:

- **A6.1+**: *agreement* (18), *agreed* (11), *agree* (6), *equality* (5), *agreements* (3), *common ground* (1).
- **S5+**: *thanks* (42), *grateful* (10), *thank* (6), *gratitude* (4), *thanking* (4), *thanked* (4), *diplomacy* (2)
- **S1.2.4+:** *thanks* (42), *grateful* (10), *thank* (6), *gratitude* (4), *thanking* (4), *thanked* (4)

Female politicians' preferences for consensus building and collaboration can also be seen from female politicians' preferences for speech acts and discussion in particular, including **Z4** ("Discourse bin"), **Q2.1** ("Speech: Communicative"), **Q2.2** ("Speech acts"), and **X3.2** ("Sensory: Sound"). Their typical tokens are as follows:

- **Z4:** *thank_you* (40), *please* (13), *well_done* (10), *right* (5), *once_again* (3), *sorry* (3), *hey* (2), *yes* (2)
- **Q2.1**: *discuss* (26), *discussing* (14), *talk* (13), *discussed* (13), *speaker* (12), *discussion* (12), *spoke* (11)
- **Q2.2**: *congratulations* (52), *welcome* (38), *condolences* (16), *call* (15), *critical* (13), *report* (8)

Therefore, female politicians tend to favor a consensual style, while male politicians tend to focus on an adversarial style. Although male politicians tend to address such tough issues as economy, female politicians address both masculine and feminine issues (Cardo, 2021).

3.5.3 Analysis of speech acts

Table 3.4 and Figure 3.3 demonstrate the distribution of self-oriented and other-oriented speech acts in the political leaders' Twitter (X). Although male and female politicians do not show much difference in their choice of self-oriented and other-oriented speech acts, it can be seen that both male and female politicians show slightly more preferences for self-oriented speech acts than other-oriented speech acts. In other words, both male and female politicians tend to be self-oriented in their use of social media.

Nevertheless, they vary in their preferences for specific speech acts. In self-oriented speech acts, female politicians show their preferences for "self-reporting moment and information" and "self-reporting mood", while male politicians show their preferences for "promoting self and others". Since "self-reporting moment and information" and "self-reporting mood" can be less face-threatening than "promoting self and others" (Liu et al., 2022),

Table 3.4 Frequencies of speech acts

Types	*Speech acts*	*Female politician*		*Male politician*	
Self-oriented	Promoting self and others	180	14%	296	24%
	Self-reporting mood	133	10%	85	7%
	Self-Reporting moment and information	415	31%	316	26%
	Subtotal	**728**	**55%**	**697**	**57%**
Other-oriented	Eliciting response	5	0%	4	0%
	Directive	167	13%	125	10%
	Greetings	80	6%	86	7%
	Showing judgment and appreciation	186	14%	230	19%
	Showing gratitude	104	8%	54	4%
	Congratulating	58	4%	34	3%
	Subtotal	**600**	**45%**	**533**	**43%**
	Total	**1328**	**100%**	**1230**	**100%**

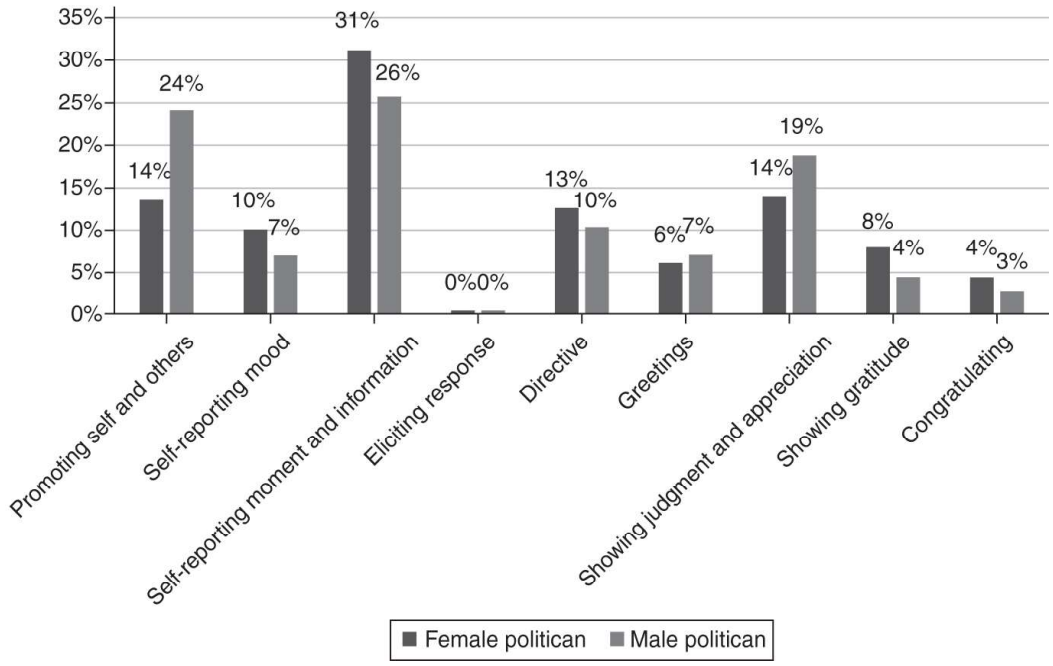


Figure 3.3 Distribution of speech acts among political leaders.

it suggests that female politicians can be less face-threatening than male politicians. Examples are as follows:

1. *Promoting self and others*: "When I signed the American Recovery and Reinvestment Act in 2009, it included $90 billion for clean energy projects. At the time, it was the *biggest investment* in renewable energy in American history." (Barack Obama)
2. *Self-reporting moment and information*: "Hey! *Sitting* outside the Pt Chev bowling club, and ready for your questions!" (AskJacinda, [Jacinda Ardern])
3. *Self-reporting mood*: "Truly heartbreaking to hear of the death of my friend Shinzo Abe, former Prime Minister of Japan killed in the most appalling of circumstances while campaigning for his political party." (Theresa May)

This suggests that male politicians are more concerned about constructing a professional image (Liu et al., 2022) and highlighting their achievements, competence, and abilities, while tweets of female politicians are characterized by the storytelling and narrative style, which can enhance the authenticity and intimacy of Twitter (X) users (Zhang & Wu, 2018). It indicates female politicians' attempt to interact and engage with the public through self-disclosure and narrative practices. Besides, female politicians make more frequent references to their mood than their male counterparts.

In the choice of other-oriented speech acts, both male and female politicians show preferences for "directives" and "showing judgment and appreciation". While "directives" are used to asking their followers to do something, "showing judgment and appreciation" is used to express opinions and comments on a certain issue. Both speech acts are face-threatening. The preference for both speech acts by both male and female politicians can be attributed to their statuses as top politicians.

Nevertheless, male politicians show their preferences for "showing judgment and appreciation", whereas female politicians show their preferences for "directives". In other words, male politicians put more emphases on constructing an authoritative identity than do female politicians. By contrast, female politicians are concerned more about interacting with their followers.

4. *Show judgment and appreciation*: "Senator Rick Scott's ultra-MAGA agenda could raise taxes on 75 million American families, over 95% of whom make less than $100,000 a year. *Their plan is going to make working families poorer*." (Joe Biden)
5. *Directives*: "As of today, abortion is still legal in the U.S. *Please share*." (Hillary Clinton)
6. *Showing gratitude*: "Thanks for the warm welcome earlier – and for all the fantastic work you do to help children and families in Govan." (Nicola Sturgeon)

This can also be seen from female politicians' greater preferences for "showing gratitude" and "congratulating" than male politicians. Both speech acts can meet the positive face need of their followers. To sum up, while both male and female politicians tend to be more self-oriented than other-oriented in their choice of speech acts. Male politicians tend to be more concerned about constructing an authoritative image, while female politicians tend to be more engaging.

3.6 Discussion and conclusion

To sum up, this study gives a corpus-assisted discourse study of the gender performances of the top ten male and female politicians on Twitter (X). Co-occurrence network analysis of the two corpora reveals the shared concerns of both male and female politicians. Nevertheless, the ranking of different themes also suggests male and female politicians' different preferences for these themes. While male politicians show their preferences for addressing domestic economic problems and the support toward Ukraine, female politicians show their preferences for international climate issues, the protection of women's rights, and the reduction of domestic violence. Key SMC analysis further reveals that male and female politicians tend to show different preferences for some key topics and communication styles. Male politicians tend to focus on the masculine issues of economy and feature an adversarial communication

style. By contrast, female politicians also show their preferences for trade and feature a consensual communication style. Analysis of speech acts further reveals that both male and female politicians show preferences for self-oriented rather than other-oriented speech acts. Nevertheless, male politicians tend to focus more on promoting self and others and showing judgment and appreciation, while female politicians tend to focus more on self-reporting moment and information, directives, and showing gratitude.

Therefore, a corpus-assisted discourse study of male and female politicians' language use on social media finds both similarities and differences. While female politicians tend to address both masculine and feminine issues in order to appear both competent and cooperative. Male politicians still highlight their competence in dealing with these tougher economic and political issues. In other words, female politicians still have to walk the fine line of the gender "double bind" (Burns & Murdie, 2018; Diekman, 2007). On the one hand, female politicians tend to underline these feminine issues, such as poverty and environment; on the other hand, they have also underlined their competence in dealing with the so-called masculine issue of trade. By contrast, male politicians tend to address the tough masculine issues of economy and security and features an adversarial communication style (Cardo, 2021; McDonnell, 2020). In other words, although female and male politicians behave in a similar way by focusing on self-oriented speech acts, female politicians do take advantage of social media to represent women and speak for women' rights and adopt a consensual communication style (Cardo, 2021). This certainly helps to feminize politics and make the public feel represented. In this sense, social media do help female politicians "bypass traditional gatekeepers and redress the representation gap between what counts as potentially meaningful and what is considered a 'minority' concern" (Cardo, 2021, p. 40). Therefore, a combination of text mining and traditional CL methods can allow us to analyze large corpora at different levels and triangulate our findings. It is hoped that this study can lead to more CADS studies on the dynamic relations between gender and political communication on social media. Nevertheless, these studies only focus on a limited sample of posts. Future studies will collect more data to verify current findings and present a more comprehensive picture of gender performances of male and female politicians on social media. Nevertheless, the gender performances can be examined at different levels of discourse from different perspectives. This study only illustrates how a corpus-assisted discourse study approach can contribute to the findings of previous studies. More detailed studies can be conducted toward this endeavor.

References

Austin, J. L. (1962). *How to Do Things with Words*. Cambridge: Harvard University Press.

Bachmann, I., Harp, D., & Loke, J. (2018). Covering Clinton (2010–2015): Meaning-making strategies in US magazine covers. *Feminist Media Studies*, *18*(5), 793–809.

Baker, P. (2005). *Public Discourses of Gay Men*. London: Routledge.

Baker, P. (2006). *Using Corpora in Discourse Analysis*. London and New York: Continuum.
Baker, P. (2008). *Sexed Texts: Language, Gender and Sexuality*. London: Equinox.
Baker, P., Gabrielatos, C., KhosraviNik, M., Krzyżanowski, M., McEnery, T., & Wodak, R. (2008). A useful methodological synergy? Combining critical discourse analysis and corpus linguistics to examine discourses of refugees and asylum seekers in the UK press. *Discourse & Society*, *19*(3), 273–306.
Bigler, R. S., & Liben, L. S. (2007). Developmental intergroup theory: Explaining and reducing children's social stereotyping and prejudice. *Current Directions in Psychological Science*, *16*(3), 162–166.
Burns, C., & Murdie, A. (2018). Female chief executives and state human rights practices: Self-fulfilling the political double bind. *Journal of Human Rights*, *17*(4), 470–484.
Butler, J. (2006). *Gender Trouble: Feminism and the Subversion of Identity*. New York: Routledge.
Cardo, V. (2021). Gender politics online? Political women and social media at election time in the United Kingdom, the United States and New Zealand. *European Journal of Communication*, *36*(1), 38–52.
Carli, L. L., & Eagly, A. H. (2001). Gender, hierarchy, and leadership: An introduction. *Journal of Social Issues*, *57*(4), 629–636.
Coates, J. (2015). *Women, Men and Language: A Sociolinguistic Account of Gender Differences in Language* (3rd ed.). London and New York: Routledge.
Diekman, A. B. (2007). Negotiating the double bind: Interpersonal and instrumental evaluations of dominance. *Sex Roles*, *56*(9–10), 551–561.
Duffy, B. E., & Pruchniewska, U. (2017). Gender and self-enterprise in the social media age: A digital double bind. *Information, Communication & Society*, *20*(6), 843–859.
Evans, H. K., & Clark, J. H. (2016). "You tweet like a girl!". *American Politics Research*, *44*(2), 326–352.
Evans, H. K., Cordova, V., & Sipole, S. (2014). Twitter style: An analysis of how house candidates used Twitter in their 2012 campaigns. *Political Science & Politics*, *47*(02), 454–462.
Everitt, J., Best, L. A., & Gaudet, D. (2016). Candidate gender, behavioral style, and willingness to vote: Support for female candidates depends on conformity to gender norms. *American Behavioral Scientist*, *60*(14), 1737–1755.
Fairclough, N. (1989). *Language and Power*. London: Longman.
Fairclough, N. (1992). *Discourse and Social Change*. Cambridge: Polity Press.
Fairclough, N. (2000). *New Labour, New Language?* London: Routledge.
Flowerdew, J. (1997). The discourse of colonial withdrawal: A case study in the creation of mythic discourse. *Discourse & Society*, *8*(4), 453–477.
Gainous, J., & Wagner, K. M. (2013). *Tweeting to Power: The Social Media Revolution in American Politics*. Oxford: Oxford University Press.
Haarman, L., & Lombardo, L. (2009). *Evaluation and Stance in War News*. New York and London: Continuum
Hardt-Mautner, G. (1995). "Only connect": Critical discourse analysis and corpus linguistics. In *UCREL Technical Paper 6*. Lancaster University.
Hayes, D. (2011). When gender and party collide: Stereotyping in candidate trait attribution. *Politics & Gender*, *7*(02), 133–165.
Heilman, M. E. (2001). Description and prescription: How gender stereotypes prevent women's ascent up the organizational ladder. *Journal of Social Issues*, *57*(4), 657–674.

Herrnson, P. S., Lay, J. C., & Stokes, A. K. (2003). Women running "as women": Candidate gender, campaign issues, and voter-targeting strategies. *The Journal of Politics*, *65*(1), 244–255.

Higuchi, K. (2016). A two-step approach to quantitative content analysis: KH-Coder tutorial using *Anne of Green Gables* (Part I). *Ritsumeikan Social Science Review*, *52*(3), 77–91.

Holmes, J., & Meyerhoff, M. (2003). Different voices, different views: An introduction to current research in language and gender. In J. Holmes & M. Meyerhoff (Eds.), *The Handbook of Language and Gender* (pp. 1–18). Malden and Oxford: Blackwell Publishing.

Huddy, L., & Terkildsen, N. (1993). Gender stereotypes and the perception of male and female candidates. *American Journal of Political Science*, *37*(1),119–147.

Jackson, N., & Lilleker, D. (2011). Microblogging, constituency service and impression management: UK MPs and the use of Twitter. *The Journal of Legislative Studies*, *17*(1), 86–105.

Jamieson, K. H. (1995). *Beyond the Double Bind: Women and Leadership*. Oxford: Oxford University Press.

Kotthoff, H., & Wodak, R. (1997). *Communicating Gender in Context*. Amsterdam and Philadelphia: John Benjamins.

Leary, M. R., & Kowalski, R. M. (1990). Impression management: A literature review and two-component model. *Psychological Bulletin*, *107*(1), 34.

Lee, J. (2013). 'You know how tough I am?' Discourse analysis of US Midwestern congresswomen's self-presentation. *Discourse & Communication*, *7*(3), 299–317.

Lee, J., & Lim, Y.-s. (2016). Gendered campaign tweets: The cases of Hillary Clinton and Donald Trump. *Public Relations Review*, *42*(5), 849–855.

L'Hôte, E. (2010). New Labour and globalization: Globalist discourse with a twist? *Discourse & Society*, *21*(4), 355–376.

Liu, M., & Huang, J. (2023). Framing responsibilities for climate change in Chinese and American newspapers: A corpus-assisted discourse study. *Journalism*. https://doi.org/10.1177/14648849231187453

Liu, M., & Jiang, C. (2019). Constant fear, but lingering nostalgia: British press representations of post-colonial Hong Kong 20 years on. *Discourse & Communication*, *13*(6), 630–646.

Liu, M., & Li, C. (2017). Competing discursive constructions of China's smog in Chinese and Anglo-American English-language newspapers: A corpus-assisted discourse study. *Discourse & Communication*, *11*(4), 386–403.

Liu, M., & Lin, L. (2021). "One country, two systems": A corpus-assisted discourse analysis of the politics of recontextualization in British, American and Chinese Newspapers. *Critical Arts*, *35*(3), 17–34.

Liu, M., & Ma, J. (2022). The politics of fear in Hong Kong protest representations: A corpus-assisted discourse study. *Journal of Language and Politics*, *21*(1), 37–59.

Liu, M., & Zhang, Y. (2018). Discursive constructions of scientific (un) certainty about the health risks of China's air pollution: A corpus-assisted discourse study. *Language & Communication*, *60*, 1–10.

Liu, M., Zhao, R., & Feng, J. (2022). Gender performances on social media: A comparative study of three top key opinion leaders in China. *Frontier in Psychology*, *13*, 1046887. https://doi.org/10.3389/fpsyg.2022.1046887

Liu, M., & Zhong, J. (2020). Between national and local: Identity representations of post-colonial Hong Kong in a local English newspaper. *Discourse, Context & Media*, *36*, 100401. https://doi.org/https://doi.org/10.1016/j.dcm.2020.100401

Marchi, A., & Taylor, C. (2009). Establishing the EU: The representation of Europe in the press in 1993 and 2005. In A. H. Jucker, D. Schreier, & M. Hundt (Eds.), *Corpora: Pragmatics and Discourse* (pp. 203–226). Switzerland: Ascona.

McDonnell, A. (2020). Clinton stated, Trump exclaimed! *Journal of Language and Politics*, *19*(1), 71–88. https://doi.org/10.1075/jlp.19085.mcd

Meeks, L. (2016). Gendered styles, gendered differences: Candidates' use of personalization and interactivity on Twitter. *Journal of Information Technology & Politics*, *13*(4), 295–310.

Melendres, C. F., & Barea, A. O. (2021). Gender inequality on Twitter during the UK election of 2019. *Quaderns de Filologia: Estudis Lingüístics*, *51*, 151–173.

Milani, T. M. (2013). Are 'queers' really 'queer'? Language, identity and same-sex desire in a South African online community. *Discourse & Society*, *24*(5), 615–633.

Morley, J., & Bayley, P. (Eds.). (2009). *Corpus-Assisted Discourse Studies on the Iraq Conflict: Wording the War*. London and New York: Routledge.

Partington, A., Morley, J., & Haarman, L. (2004). *Corpora and Discourse*. Bern: Peter Lang.

Sanbonmatsu, K. (2002). Gender stereotypes and vote choice. *American Journal of Political Science*, *46*(1), 20–34.

Statista. (2023). *Number of X(formerly Twitter) users worldwide from 2019 to 2024*. Retrieved November 22 from www.statista.com/statistics/303681/twitter-users-worldwide/

Stubbs, M. (1996). *Text and Corpus Analysis: Computer-assisted Studies of Language and Culture*. Blackwell.

Trent, J. S., & Friedenberg, R. V. (1991). *Political Campaign Communication*. London: Rowman & Littlefield.

Wang, G., & Ma, X. (2020). Representations of LGBTQ+ issues in China in its official English-language media: a corpus-assisted critical discourse study. *Critical Discourse Studies*, *18*(2), 1–19.

Windsor, L., Mitchell, S. M., Osborn, T., Dietrich, B., & Hampton, A. J. (2022). Gender, language, and representation in the United States Senate. *Journal of Language and Politics*, *21*(6), 919–943. https://doi.org/10.1075/jlp.21053.win

Wodak, R. (1997). *Gender and Discourse*. London: Sage.

Wodak, R. (2016). The discourse-historical approach. In R. Wodak & M. Meyer (Eds.), *Methods of Critical Discourse Analysis* (3rd ed., pp. 23–61). London: Sage.

Wodak, R., de Cillia, R., Reisigl, M., & Liebhart, K. (2009). *The Discursive Construction of National Identity* (2nd ed.). Edinburgh: Edinburgh University Press.

Wodak, R., & Meyer, M. (2016). *Methods of Critical Discourse Analysis*. London: Sage.

Yarchi, M., & Samuel-Azran, T. (2018). Women politicians are more engaging: male versus female politicians' ability to generate users' engagement on social media during an election campaign. *Information, Communication & Society*, *21*(7), 978–995.

Zhang, M., & Wu, D. D. (2018). A cross-cultural analysis of celebrity practice in microblogging. *East Asian Pragmatics*, *3*(2), 179–200.

Zhu, H. (2016). Searchable talk as discourse practice on the internet: The case of "# bindersfullofwomen". *Discourse, Context & Media*, *12*, 87–98.